Assignment Brief: COMP2431 OO Design & Development 2025-26 Semester 1+2

Assignment number 2	'Viva style' Team Presentation
Word Limit or equivalent (e.g. time)	Portfolio & Demonstration
Weighting	60%
Learning Outcomes Assessed	2. Design appropriate object-oriented solutions to problems. 3. Construct object-oriented software components for a given design, employing a modern object-oriented language. 4. Produce appropriate design and development documentation for object-oriented software.
Submission date	Team Presentation schedule to be released in Week 22, (WC 13th April 2026). The Team presentations & VIVAs will take place during week 24 (WC 27th April 2026) If required additional demonstrations will be held in Revision Week (WC 4th May 2026) Team documentation submit by 1500 (3pm) Thursday 7th May 2026
Feedback date	All assignment feedback will be issued on the 20th working day following the submission deadline. Feedback will be released on: 5th June 2026
Module Leader	Bradley Carwardine b.carwardine@worc.ac.uk
Verified by	Chris Blythe

If anything about this assignment is not clear to you, please contact your module leader.

What do I need to do to make a success of this assignment?	You will compete assignment 2 in teams of 3-4. Your team will create an extensive **Design Portfolio** and coded **Solution Artefact** to a given brief from a client. The client is Gino's Cafeteria. The client brief document (**Gino's Cafeteria Brief**) detailing the client requirements can be found on Blackboard in Assessment 2. **Design Portfolio** The team will produce: • A clear analysis of the software problems, (requirements), that have been selected from **Gino's Cafeteria Brief** that need to be solved. (See Grading Methodology) • Design documents should be created for each requirement developed. Design documents could include, but is not limited to: ○ Use case descriptors/diagram ○ Class Discovery methodology/CRC Cards ○ UML Diagram ○ Sequence diagrams • A detailed explanation of how your software meets the chosen requirements identified in the **Gino's Cafeteria Brief**. **Solution Artefact** The team will create: • A single solution artefact written in JAVA. • The artefact created should demonstrate object-oriented methodologies have been used when writing the code of the artefact, including but not limited to: • Abstract classes • Interfaces • Collections: Arrays/Array Lists • File handling • Exception handling • The use of SOLID Design Principles being employed should be evident within the code. • A clear attempt to achieve loose coupling and strong cohesion should be demonstrated. • Documented evidence of testing and debugging. • A 'code-wiki' or commented code listings that documents the functional elements in the code. **Presentation of Design Portfolio and Demonstration of Artefact** Your team will present the Design Portfolio and Demonstrate the Artifact created. Within the presentation and demonstration, the following will be completed: • A review of the produced design. This will be to 'Pitch' how the completed design meets the requirements in the client brief. • The pitch will not be a one way 'presentation'. The team will need to answer questions from the assessor as to how their design choices have been realised in the Solution Artefact. • An interactive 'walk-through' demonstration of the functions of the

	Solution Artefact. • During the demonstration the team will need to answering questions from the assessor relating to the code base of the Artefact as they arise. **Grading Methodology** In addition to the quality of response to each section of the assessment the following should be used to help your team to target a desired grade. To gain a **Passing grade each team member** must take responsibility for the design, coding & testing of **1 Functional Requirement** each, and **w**orking as **a team** assist in the design, development and testing of **2 Advanced Requirements.** The team should work together to integrate all the requirements developed into one application. To gain a **Higher grade each team member** must take responsibility for the design, coding & testing of **2 Functional Requirements** each**,** and working as **a team** assist in the design, development and testing of **3 Advanced Requirements.** The team should work together to integrate all the requirements developed into one application. To gain the **Top grades each team member** must take responsibility for the design, coding & testing of **3 Functional Requirements** each**,** and working as **a team** assist in the design, development and testing of **4 Advanced Requirements.** The team should work together to integrate all the requirements developed into one application. **VIVA:** Each team member will answer **Individual Questions** relating to the design, coding & testing of Functional Requirements they were responsible for. **Please Note:** It is not possible to pass this assignment without designing, writing and testing code, and then being able to explain how the developed code functions. **Submission of Teamwork files** • All Teamwork files need to be collected into a Zip file. At the end of the Presentation and Vivas time will be given to submit the zip file via a Blackboard Dropbox.

How should I present my work? (delete whole section if 100% exams)	Your work must be word-processed/typed and should clearly show the student numbers of each team member. The Artefact code should be scripted IntelliJ Idea. Remember: If you copy code that has not been written by yourself, it is imperative that you reference it clearly and carefully in within your submission documentation. Failure to do so may trigger an academic misconduct investigation. The result of which may affect your overall module grade. All written work should use appropriate citations and references using the University Harvard referencing system - support is available through the library www.worc.ac.uk/library/guides/study-skills/referencing

How can I obtain guidance on my assignment?	Further details for this assignment are provided in the 'detailed assignment specifications' folder, located on Blackboard: under 'Assessment 2'. Specific sessions are timetabled for formal explanation and guidance on the assessment – please refer to the module outline. However, there will be opportunities for 1-1 guidance on the overall structure and direction of your assignment during most sessions - if you are unsure of anything, please do ask your tutor at these times or contact them by email. **AFTER** checking Blackboard for details, ensure you contact the module leader if you have any questions. Refer to the Grading Matrix below in this Assignment Brief for guidance on grading criteria

How and when do I hand my assignment in? *(Delete whole section if 100% exams)*	The team Design Portfolio, Artefact code and presentation materials should be submitted in a single zip file at the end of your presentation and viva. The file should be named using the following convention: ***TeamName* COMP2431 Design Portfolio and Artefact code.zip** Submitting the team material is required for moderation purposes. You should submit your teams work in the '**Assignment 2 Dropbox**' on Blackboard, which is available via MyDay. **You are required to keep a copy of work handed in.** See the separate Assignment Support Information document on Blackboard for help on how to submit or what to do if you are having trouble submitting your assignment.

How will my assignment be marked?	Specific marking criteria for your assignment is provided in the Grading Matrix at the end of this document. You are strongly advised to check your completed work against the Grading Matrix to ensure have completed all areas required when creating the software artefact and preparing for the presentation and viva. **Grades will be calculated from responses given during the presentation and demonstration!** You should also ensure you adhere to the word limit / word count stated in your assessment brief document, details of which can be found in the Universities Assessment Policy http://www.worc.ac.uk/aqu/documents/AssessmentPolicy.pdf

Use of Generative Artificial Intelligence (Gen AI)	We are very aware of Generative Artificial Intelligence (AI) tools, such as ChatGPT, and how tempting it could be to use it to help with assignments. We are very aware of some students using these tools, and we are very good at identifying when Generative AI tools have been used in assignments. The use of Generative AI tools to produce your assignments, either wholly or in part, is classed as academic misconduct and will result in academic misconduct proceedings being brought unless the use of Generative AI is expressly permitted by your Module Leader. This could result in you being required to retake the assessment item, the module, or even failure of your course. If you're unsure and want some advice, please contact your module tutor, module leader or personal academic tutor for guidance.

University of Worcester

Grading Matrix

This matrix captures the assessment criteria for this part of the coursework.

Student Name/Number:		**Assignment No:**	2	**Weighting:**	60%
Module Code:	COMP2431	**Assignment Title:**	**'Viva-Style' Presentation Portfolio & Demonstration**		
Module Title:	Object Oriented Design & Development	**Semester:**	2		
Learning Outcomes being assessed:					
2. Design appropriate object-oriented solutions to problems. 3. Construct object-oriented software components for a given design, employing a modern object-oriented language. 4. Produce appropriate design and development documentation for object-oriented software.					

Assessment Criteria

GRADE	Presentation of Design Portfolio (15)	Demonstration of Solution Artefact (30)	Code Documentation (15)	Individual VIVA questions (30)	SPaG - Development Evidence Document (10)
A	There was Exceptional Analysis of the problems to be solved. All the specified design documents were produced, these were created to an exceptional standard. There was an Exceptional design 'Pitch' in the presentation of how the proposed design solves the problems identified. Commensurate with this grade the required number of Functional, Extended and Non-Functional requirements were fully addressed.	The artefact was exceptionally created in JAVA The artefact demonstrated exceptional application of Object-oriented methodologies. All SOLID design principles have been utilised in an exceptional manor. Commensurate with this grade the required number of Functional, Extended and Non-Functional requirements were fully represented in the artefact. All functionality executed fully and as expected.	Excellent code documentation was produced. There was clear evidence of a full range of testing and debugging conducted. All code was commented to an excellent standard. Excellent implementation and use of a code documentation repository was demonstrated.	Exceptional breadth and depth of knowledge was demonstrated in the answers given. Clear independent thought and critically evaluative understanding was shown.	Excellent Development Evidence Document demonstrating clarity, and coherence that is well judged through high standards of written English/ spelling, punctuation and grammar used accurately to ensure ideas are communicated effectively
B	There was strong	The artefact was created in	Very good documentation	Very good breadth	Very good Development

	analysis of the problems to be solved. All the specified design documents were produced, these were created to a Strong standard. There was a Strong design 'Pitch' in the presentation of how the proposed design solves the problems identified. Commensurate with this grade the required number of Functional, Extended and Non-Functional requirements were addressed well.	JAVA to a strong standard. The artefact demonstrated very good application of Object-oriented methodologies. Most SOLID design principles have been utilised in a strong manor. Commensurate with this grade the required number of Functional, Extended and Non-Functional requirements were represented in the artefact well. All functionality executed fully and as expected.	was produced. There was clear evidence of a good range of testing and debugging conducted. All code was commented to an very good standard. A very good implementation and use of a code documentation repository was demonstrated.	and depth of knowledge was demonstrated in the answers given. Good independent thought and critically evaluative understanding was shown.	Evidence Document demonstrating clarity and coherence through very good standards of written English/ spelling, punctuation and grammar used accurately to ensure ideas are communicated effectively

C	There was a good analysis of the problems to be solved. Most of the specified design documents were produced, of the documents produced they were created to a good standard. There was a good design 'Pitch' in the presentation of how the proposed design solves the problems identified. Commensurate with this grade the required number of Functional, Extended and Non-Functional requirements were addressed.	The artefact was created in JAVA to a sound standard. The artefact demonstrated good application of a range of Object-oriented methodologies. Some SOLID design principles have been utilised in a sound manor. Commensurate with this grade the required number of Functional, Extended and Non-Functional requirements were represented in the artefact. All functionality executed and mostly as expected.	Good documentation was produced. There was good evidence of testing and debugging being conducted. All code was commented to a good standard. An implementation and use of a code documentation repository was demonstrated.	Good breadth and depth of knowledge was demonstrated in the answers given. Emerging independent thought and critically evaluative understanding was shown.	Good clear Development Evidence Document perhaps with some minor weaknesses through good standards of written English/ spelling, punctuation and grammar used accurately to ensure ideas are communicated effectively

D	There was an adequate analysis of the problems to be solved. There were some design documents produced, of the documents produced they were created to an adequate standard. There was an adequate design 'Pitch' in the presentation of how the proposed design solves the problems identified. Commensurate with this grade the required number of Functional, Extended and Non-Functional requirements were adequately addressed.	The artefact was created in JAVA to an adequate standard. The artefact demonstrated a mostly competent application of Object-oriented methodologies. A few SOLID design principles were utilised in an adequate manor. Commensurate with this grade the required number of Functional, Extended and Non-Functional requirements were represented in the artefact. The functionality mostly executed, however some functions did not execute as expected.	Adequate documentation was produced. There was evidence of some testing and debugging being conducted. Most code was commented to an adequate standard. An attempt to implementation and use a code documentation repository was demonstrated.	Sufficient breadth and depth of knowledge was demonstrated in the answers given. Little independent thought and critically evaluative understanding was shown.	Competent Development Evidence Document with some weaknesses through competent standards of written English/ spelling, punctuation and grammar used accurately to ensure ideas are communicated effectively

E-F	There was an inadequate analysis produced of the problems to be solved. There were few design documents produced, of the documents produced they were created to an inadequate standard. There was an inadequate design 'Pitch' in the presentation of how the proposed design solves the problems identified. Some of the required Functional, Extended and Non-Functional requirements have not been addressed.	The artefact was created in JAVA to an inadequate standard. The artefact demonstrated a weak implementation of Object-oriented methodologies. Very few SOLID design principles have been utilised, these have been implemented in an inadequate manor. The artefact may not have executed, and/or most functions did not execute as expected.	Inadequate documentation was produced. There was little evidence of testing and debugging being conducted. Some code was commented. A poor attempt at the implementation and use of a code documentation repository was demonstrated.	Weak breadth and depth of knowledge was demonstrated in the answers given. No independent thought and critically evaluative understanding was shown	Communication of information is inaccurate, incomplete or otherwise problematic in conveying understanding. Errors in English/spelling, punctuation and grammar which impede understanding of content

G	There was a weak analysis produced of the problems to be solved. There were very few design documents produced, of the documents produced they were created to an weak standard. There was a weak design 'Pitch' in the presentation of how the proposed design solves the problems identified. Most of the required Functional, Extended and Non-Functional requirements have not addressed.	The artefact was created in JAVA to a very poor standard. The artefact demonstrated little implementation of Object-oriented methodologies. SOLID design principles do not appear to have been utilised. The artefact failed to executed, and functions did not execute as expected.	Weak documentation was produced. There was no evidence of testing and debugging being conducted. Very little code was commented/poor commenting was produced. No attempt at the implementation and use of a code documentation repository was demonstrated.	Weak responses were given indicating very poor knowledge and understanding	Very poor communication indicating incoherence and/or seriously incomplete understanding. Many errors in English/spelling, punctuation and grammar which impede understanding of content
H / NS	Seriously inadequate or insufficient response or non-submission.	Seriously inadequate or insufficient response or non-submission.	Seriously inadequate or insufficient response or non-submission.	Seriously inadequate or insufficient response or non-submission.	Seriously inadequate or insufficient response or non-submission.

NB: RESULTS ARE PROVISIONAL UNTIL AGREED BY THE BOARD OF EXAMINERS